Package 'DTP'

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Type Package

Title Double twopiece distributions

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Description Density, distribution function, quantile function and random generation for Double two-piece distributions with 3 parameterisations: two-piece (tp), epsilon-skew (eps), and inverse scale factors (isf).

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Encoding UTF-8

LazyData true

RoxygenNote 7.3.3

Contents

`ddtp` *Probability Density Function for DTP distributions*

Description

Probability Density Function for DTP distributions

Usage

```
ddtp(x, mu, par1, par2, delta1, delta2, f, param = "tp", log = FALSE)
```

Arguments

`log`	log.p: logical; if TRUE, probabilities p are given as log(p).
`x:`	vector of quantiles.
`p:`	vector of probabilities.
`n:`	number of observations. If length(n) > 1, the length is taken to be the number required.
`mu:`	location parameter.
`par1:`	scale parameter 1.
`par2:`	scale parameter 2.
`delta1:`	shape parameter 1.
`delta2:`	shape parameter 2.
`F`	qF, rF, f: distribution function, quantile function, random function and density function associated of a symmetric random variable.
`param:`	parameterisations used.

`pdtp` *Cumulative Probability Function for DTP distributions*

Description

Cumulative Probability Function for DTP distributions

Usage

```
pdtp(x, mu, par1, par2, delta1, delta2, F, f, param = "tp", log.p = FALSE)
```

Arguments

`F`	qF, rF, f: distribution function, quantile function, random function and density function associated of a symmetric random variable.
`x:`	vector of quantiles.
`p:`	vector of probabilities.
`n:`	number of observations. If length(n) > 1, the length is taken to be the number required.
`mu:`	location parameter.
`par1:`	scale parameter 1.
`par2:`	scale parameter 2.
`delta1:`	shape parameter 1.
`delta2:`	shape parameter 2.
`param:`	parameterisations used.
`log`	log.p: logical; if TRUE, probabilities p are given as log(p).

`qdtp` *Quantile Function for DTP distributions*

Description

Quantile Function for DTP distributions

Usage

```
qdtp(p, mu, par1, par2, delta1, delta2, qF, f, param = "tp")
```

Arguments

`x:`	vector of quantiles.
`p:`	vector of probabilities.
`n:`	number of observations. If length(n) > 1, the length is taken to be the number required.
`mu:`	location parameter.
`par1:`	scale parameter 1.
`par2:`	scale parameter 2.
`delta1:`	shape parameter 1.
`delta2:`	shape parameter 2.
`F`	qF, rF, f: distribution function, quantile function, random function and density function associated of a symmetric random variable.
`param:`	parameterisations used.
`log`	log.p: logical; if TRUE, probabilities p are given as log(p).

`rdtp` *Random Number Generation for DTP distributions*

Description

Random Number Generation for DTP distributions

Usage

```
rdtp(n, mu, par1, par2, delta1, delta2, rF, f, param = "tp")
```

Arguments

`x:`	vector of quantiles.
`p:`	vector of probabilities.
`n:`	number of observations. If length(n) > 1, the length is taken to be the number required.
`mu:`	location parameter.
`par1:`	scale parameter 1.
`par2:`	scale parameter 2.
`delta1:`	shape parameter 1.
`delta2:`	shape parameter 2.
`F`	qF, rF, f: distribution function, quantile function, random function and density function associated of a symmetric random variable.
`param:`	parameterisations used.
`log`	log.p: logical; if TRUE, probabilities p are given as log(p).

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